Group 13 American Sign Language Interpretation Using Deep Learning

Jaron Zhu, Matthew Mitchell, Franky Liu
University of California, San Diego

Abstract

American Sign Language (ASL) recognition is an important step towards improving accessibility and communication for the Deaf and hard of hearing community. Automatic recognition of ASL from video remains challenging due to complex hand shapes, rapid motion, and large vocabularies of possible signs. Only a small fraction of the hearing community understands ASL thus creating a communication barrier between the Deaf and hearing community. In this work, we develop a deep learning based approach for ASL gloss classification from video clips. Our method processes short video segments by extracting spatial features from each frame using a pretrained convolutional neural network (CNN) backbone and modeling temporal dynamics with a Long Short Term Memory (LSTM) network. The model is trained on the Word-Level American Sign Language (WLASL) dataset containing 2000 unique glosses, with label smoothing used to improve generalization across classes. We analyze the impact of signer-based dataset splits, region-of-interest cropping using bounding boxes and temporal modeling. Results show that purely spatial CNN models achieve high training accuracy while suffering from overfitting. Incorporating a hybrid approach with LSTM layers improves generalization performance on unseen signers. These findings highlight the importance of temporal representation learning in sign language recognition systems.

I. INTRODUCTION

Automatic recognition of American Sign Language (ASL) is an important problem in computer vision and machine learning because it can help reduce communication barriers for the Deaf and hard of hearing community. American Sign Language is a visual language that relies on hand shape, movement, orientation and spatial positioning. Despite the widespread use of ASL, many digital platforms and services are primarily designed for spoken or written language, which limits accessibility for ASL users. Developing systems that can automatically interpret ASL from video could enable real-time translation tools, improve accessibility in educational content and support more inclusive human computer interaction.

ASL recognition is a complex spatial temporal learning task, with each sign unfolding across multiple video frames and involving rapid hand movements. In addition, large ASL datasets often contain thousands of unique glosses, (word-level labels), which makes the task a large multi-class classification problem. These challenges motivate the use of deep learning models capable of learning both spatial and temporal patterns from video data.

In this capstone project, we develop a deep learning approach for ASL gloss recognition from video clips. The input to our algorithm is a short sequence of video frames extracted from an ASL signing clip. The sequences have a spatial dimension of 224x224 pixels and capture the motion of a signer performing a single word or gloss. The model first uses convolutional neural network (CNN) to extract spatial features from each frame, and then Long Short Term Memory (LSTM) models capture the motional temporal sequence of the sign. The output of the model is a predicted ASL gloss from a vocabulary of 2000 unique glosses. By combining spatial feature extraction with temporal sequence modeling, this approach aims to effectively capture the visual and motion patterns necessary for recognizing signs from videos.

II. RELATED WORK

Sign language recognition has received significant attention in the computer vision community. Earlier approaches relied on handcrafted features such as motion descriptors

or skeletal tracking data extracted from video frames. However, these methods often struggled to generalize across different lighting conditions, signer styles, and backgrounds.

Recent work has shifted toward deep learning approaches. CNNs are widely used to extract spatial features from images, while recurrent neural networks (RNNs) and LSTM networks capture temporal dependencies across video frames. Other architectures such as 3D convolutional networks and graph convolution networks have also been proposed for modeling motion patterns.

The Word-Level American Sign Language (WLASL) dataset has become a widely used benchmark for evaluating sign language recognition algorithms. It contains thousands of labeled video clips across many different signers and gloss categories, making it well suited for evaluating generalization performance.

This dataset was used by Dongxu Li and team[1], where they approached the dataset using both appearance and pose based models, such as CNN, RNN, I3D, GRU, and convolutional networks. This approach is especially similar to our project, with the difference being in that we used a much lighter pipeline that consisted of only CNN and LSTM.

Their approach consisted of extracting features from video frames using CNN similar to our methods, but they applied 3D networks and classified signs with a soft max over the predicted glossary.

Table 3: Top-1, top-5, top-10 accuracy (%) achieved by each model (by row) on the four WLASL subsets.

Method	WLASL100			WLASL300			WLASL1000			WLASL2000		
	top-1	top-5	top-10	top-1	top-5	top-10	top-1	top-5	top-10	top-1	top-5	top-10
Pose-GRU	46.51	76.74	85.66	33.68	64.37	76.05	30.01	58.42	70.15	22.54	49.81	61.38
Pose-TGCN	55.43	78.68	87.60	38.32	67.51	79.64	34.86	61.73	71.91	23.65	51.75	62.24
VGG-GRU	25.97	55.04	63.95	19.31	46.56	61.08	14.66	37.31	49.36	8.44	23.58	32.58
I3D	**65.89**	**84.11**	**89.92**	**56.14**	**79.94**	**86.98**	**47.33**	**76.44**	**84.33**	**32.48**	**57.31**	**66.31**

Analyzing the results from their research, we can see that the best accuracy for top-1 is around 66%. We will show our accuracies and the methods of achieving it in later sections to form comparisons.

Dongxu and team also noted some improvements that could have been made to improve model variability and accuracy; such as constructing a more precise region of interest, which we have taken note of and implemented bounding boxes on the region of interest in our model.

III. DATASET AND FEATURES

Our experiments use the WLASL dataset, a large-scale benchmark dataset for sign language recognition introduced by Li et al. The dataset contains 21,083 labeled video clips representing 2,000 unique ASL glosses performed by 119 different signers. Each video clip corresponds to a signer performing a single word-level sign and including metadata such as the gloss label, signer identity, clip identifier, and bounding box coordinates that localize the signer within each frame.

The dataset is split into training, validation, and test sets using signer-based partitions, ensuring that individual signers appear in only one subset of the data. This prevents the model from learning signer-specific visual characteristics rather than the gesture itself and provides a more realistic evaluation of generalization performance.

Each video is processed into a fixed-length sequence of frames to represent the temporal evolution of the sign gesture. Frames are sampled uniformly across the duration of the clip so that both the beginning and end of the gesture are represented. If a video contains more frames than the target sequence length, frames are downsampled; if it contains fewer frames, frames are padded or repeated to maintain consistent dimensionality. All frames are resized to 224 × 224 pixels, which matches the input resolution expected by the ResNet backbone used in our model.

To reduce background noise and apply focus on the model of relevant motion on the signer's hands and upper body, we apply region-of-interest (ROI) cropping using CNN. The CNN learns to focus on the most relevant visual regions in each frame, such as the hands, arms and upper body of the signer. Through the

use of convolutional filters and fully connected layers, the network progressively emphasizes these regions while suppressing background noise. If the model fails to apply an CNN on the video frames, a fallback center crop method is applied by applying a center crop of 192x192 pixels onto the frames. An example of the preprocessing pipeline and resulting cropped frames is shown in Fig. 1. When the CNN model analyzes the frames and finds an area where motion occurs, it returns a bounding box on the region of interest. Then a cropping function is applied to extract only the region of interest and removes background noise, keeping only the signer's hand.

The WLASL dataset supports several potential predictive tasks, including word-level sign classification, signer identification, and gesture segmentation within continuous video streams. In this project we focus on word-level gloss classification, where the goal is to predict the corresponding ASL gloss given a sequence of video frames representing a single sign. This task aligns directly with the dataset annotations and serves as a fundamental step toward automatic sign language translation systems.

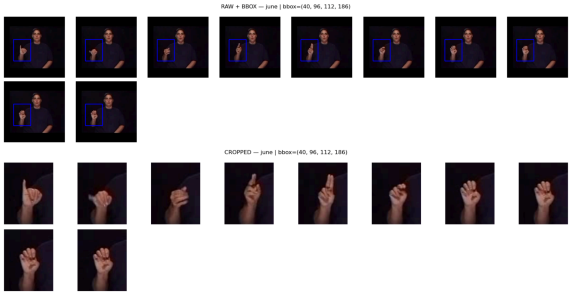

Fig. 1. Example frame sequence from the WLASL dataset illustrating region-of-interest preprocessing. The top row shows original video frames with the CNN model highlighting the signer's hand region, while the bottom row shows the corresponding cropped hand regions zoomed in and used as input to the model.

The input to our model therefore consists of a temporal sequence of RGB image frames representing the motion of a signer performing a sign. Spatial features such as hand shape and position are extracted from each frame using a convolutional neural network. These frame-level representations form the primary features used by the model to learn the visual and temporal patterns associated with different ASL signs.

V. METHODS

A. Problem Formulation

The goal of this project is to classify American Sign Language (ASL) signs from short video clips. Each clip represents a signer performing a single word-level sign, referred to as a gloss. After preprocessing, each video is represented as a sequence of T frames:

$$X = \{x_1, x_2, \ldots, x_T\}$$

where each frame x_T is an RGB image resized to 224×224 pixels. The objective of the model is to predict the corresponding gloss label:

$$y \in \{1, 2, \ldots, C\}$$

where C is the number of possible ASL gloss classes. This task can therefore be framed as a sequence classification problem in which the model learns a function:

$$f(X) = \hat{y}$$

that maps a sequence of frames to a predicted gloss label. While ASL gesture contains both visual and motion information, the model must capture spatial structure within individual frames as well as temporal relationships between frames.

To address this problem, we evaluate two architectures: a CNN baseline that focuses on spatial information and applying an LSTM model that incorporates temporal modeling. This allows us to use an hybrid architecture of an CNN-LSTM approach, with CNN extracting spatial features from frames and the LSTM network to model temporal sequence of hand sign.

B. Spatial Feature Extraction Using CNNs

Convolutional neural networks (CNNs) are widely used for image recognition tasks because they automatically learn hierarchical visual features directly from pixel inputs. Early convolutional layers typically capture low-level patterns such as edges and textures, while deeper layers capture more complex structures such as object shapes and spatial relationships.

In the context of ASL recognition, CNNs learn visual cues related to hand shape, orientation, and body posture. These features are important because many signs can already be partially distinguished from static visual information. For each frame x_T, the CNN produces a feature representation:

$$z_T = \phi(x_T)$$

where $\phi(\cdot)$ represents the feature extraction process. These feature vectors summarize the important spatial information contained in each frame and serve as the input to the classification stage.

C. ResNet-50 Backbone

To obtain high-quality visual features, our model uses ResNet-50 as the CNN backbone. ResNet is a deep convolutional network architecture that introduces residual connections, allowing information to flow more easily through multiple network layers. These connections help stabilize training and improve performance in deep networks.

In practice, the pretrained ResNet-50 network processes each frame of the video independently and produces a compact feature vector representing the visual structure of that frame. These features capture important information such as hand configuration, orientation, and the relative position of the signer's arms.

These frame-level feature vectors are then used as the input to the classification model.

D. CNN Baseline Model

The first model evaluated in this project is a CNN-based classifier. In this approach, spatial features extracted from video frames are used directly to predict the ASL gloss label. Given a feature representation z, the classifier produces scores for each possible class. These scores are converted into probabilities using the softmax function:

$$P(y = c|z) = \frac{exp(o_c)}{\sum_{j=1}^{C} exp(o_j)}$$

where o_c is the score for class c. The predicted class is the one with the highest probability.

This baseline model allows us to evaluate how well spatial information alone can distinguish different ASL signs. While static visual cues can be informative, this approach does not explicitly model the motion of the gesture across frames.

E. Temporal Modeling with LSTM

Many ASL gestures depend on motion patterns rather than static hand shapes. To capture these temporal relationships, we extend the baseline model by incorporating a Long Short-Term Memory (LSTM) network.

LSTMs are a type of recurrent neural network designed to process sequential data while maintaining a memory of previous inputs. This allows the model to retain information from earlier frames and update its internal representation as new frames are observed.

In our architecture, the sequence of frame features:

$$(z_1, z_2, \ldots, z_T)$$

is passed to the LSTM in chronological order. As the sequence progresses, the LSTM updates its hidden state to capture how the gesture evolves over time.

The final hidden state of the LSTM provides a representation of the entire video sequence and is used as the input to the final classification layer.

F. CNN–LSTM Architecture

The full CNN–LSTM model combines spatial feature extraction with temporal modeling. Each video frame is first processed by the ResNet backbone to generate a feature vector. These vectors are then passed sequentially through the LSTM network, which learns temporal dependencies between frames.

The final LSTM representation is then passed to two fully connected layers that produce class probabilities through the softmax activation function.

This architecture allows the model to capture both spatial and temporal information. The CNN component learns visual patterns within each frame, while the LSTM models the motion patterns that occur across the video sequence. This combination is well suited for ASL recognition, where gestures are defined by both hand configuration and movement.

G. Training Objective

The model is trained using cross-entropy loss, which measures the difference between the predicted probability distribution and the true class label. For a single training example, the loss is defined as

$$L = -\log P(y|X)$$

Minimizing this loss encourages the model to assign higher probability to the correct gloss label. During training, gradients are computed through both the CNN and LSTM components, allowing the model to jointly learn spatial and temporal representations.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup

To evaluate the effectiveness of the proposed approach for American Sign Language recognition, experiments were conducted using the WLASL dataset. Each sample consists of a temporal sequence of RGB frames extracted from a video clip representing a single ASL gloss. The frames were resized to 224 × 224 pixels to match the input resolution required by the convolutional neural network backbone used in the model.

Two primary architectures were evaluated. The baseline model uses a convolutional neural network (CNN) to extract spatial features from individual frames. Each frame is processed independently through a pretrained ResNet-50 network to produce a feature representation describing visual properties such as hand shape, position, and orientation. To capture temporal relationships between frames, a hybrid CNN–LSTM architecture was also implemented. In this configuration, frame-level features extracted by the CNN are passed to a Long Short-Term Memory (LSTM) network that models temporal dependencies across the sequence of frames. The final hidden state of the LSTM is then passed to a fully connected classification layer followed by a softmax activation to generate probabilities over the ASL gloss vocabulary.

Models were trained using the AdamW optimizer, which provides adaptive gradient updates while incorporating decoupled weight decay to improve generalization. Training was performed using mini-batches of frame sequences, allowing efficient updates of model parameters while maintaining stable gradient estimates.

Two dataset partitioning strategies were evaluated. The first uses a random split in which samples are randomly divided into training and validation sets. The second uses a signer-based split in which each signer appears in only one partition of the dataset. The signer-based split provides a more realistic evaluation because it prevents the model from learning signer-specific visual characteristics. In addition, experiments

were conducted both with and without bounding box region-of-interest (ROI) preprocessing. ROI cropping uses bounding box annotations to focus the input frames on the signer's hands and upper body, reducing irrelevant background information.

B. Evaluation Metrics

The task considered in this work is a multi-class classification problem in which the model predicts the correct ASL gloss given a sequence of video frames. Performance was therefore evaluated using classification accuracy.

Top-1 accuracy measures the proportion of samples in which the model's highest probability prediction matches the ground truth label. This metric is defined as

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} 1(y_i = \hat{y}_i)$$

where y_i represents the true label, $\hat{y}_i$ represents the predicted label, and N is the total number of samples. Accuracy provides a straightforward measure of how often the model correctly identifies the target sign.

In addition to validation accuracy, training accuracy was monitored to evaluate potential overfitting. Large differences between training accuracy and validation accuracy indicate that the model may be memorizing training samples rather than learning patterns that generalize to new data.

C. Model Performance

The results of the evaluated models are summarized in Table I. The table compares performance across different model architectures, dataset splitting strategies, and preprocessing configurations.

The CNN model trained using a random dataset split achieves the highest training accuracy, reaching 97.9%. However, its validation accuracy is significantly lower at 60.5%, indicating that the model may be overfitting to the training data. This behavior is largely caused by signer leakage, where the same signer appears in both the training and validation sets. In such cases, the model may learn visual characteristics associated with specific individuals rather than learning the gesture itself.

When the dataset is partitioned using signer-based splits, validation accuracy decreases to approximately 52–54%. Although this reduction may initially appear unfavorable, it represents a more realistic evaluation of model performance because the model must recognize gestures performed by previously unseen individuals.

The effect of bounding box preprocessing can also be observed in Table I. Applying ROI cropping slightly improves validation accuracy for the CNN model, increasing performance from 52.8% to 53.6%. By focusing the input on the signer's hands and upper body, ROI cropping reduces background noise and allows the model to concentrate on the most informative visual features.

Among the evaluated models, the CNN–LSTM architecture demonstrates the strongest generalization behavior. Although its training accuracy is lower at 91.6%, the validation accuracy remains competitive at 54.2% while reducing the gap between training and validation performance. This indicates that incorporating temporal modeling helps the model capture motion patterns across frames rather than relying solely on static spatial features.

TABLE 1. MODEL PERFORMANCE

Model	Split	BBox	Train Acc	Val Top-1
CNN	Random	No	97.9%	60.5%
CNN	Signer	No	96.8%	52.8%
CNN	Signer	Yes	96.9%	53.6%
CNN-LSTM	Signer	Yes	91.6%	54.2%

D. Analysis of Model Behavior

The experimental results highlight several important aspects of sign language recognition using deep learning. First, the high training accuracy achieved by CNN models suggests that spatial information alone can capture many visual characteristics of ASL gestures, such as hand shape and orientation. However, many signs differ primarily in their motion patterns rather than static appearance.

The CNN–LSTM architecture addresses this limitation by modeling temporal relationships between frames. By learning how hand positions change over time, the LSTM layer captures motion patterns that distinguish gestures that may appear similar in individual frames. This temporal modeling helps improve generalization to new examples.

The comparison between random and signer-based dataset splits further demonstrates the importance of proper evaluation procedures in sign language recognition research. Random splits may produce artificially inflated performance because the model can exploit signer-specific visual cues rather than learning the gesture itself.

Finally, the improvements observed with ROI preprocessing suggest that focusing the model on the most relevant region of the frame improves classification performance. Removing unnecessary background information helps the model learn more robust spatial representations of the sign gesture.

E. Limitations and Error Analysis

Despite the promising results obtained in this study, several challenges remain. One limitation is the class imbalance present in the WLASL dataset, where certain glosses appear far more frequently than others. This imbalance can bias the model toward predicting common signs while reducing performance on less frequent gestures.

Another challenge arises from the visual similarity between certain ASL signs. Some gestures share similar hand shapes and differ only in subtle motion patterns or orientation changes. Distinguishing these gestures requires highly precise temporal modeling.

Future work could address these challenges by incorporating class-weighted loss functions, exploring more advanced video recognition architectures such as 3D convolutional networks or transformer-based models, and conducting more detailed analysis using confusion matrices and error visualizations.

VII. CONCLUSION

This project explored the use of deep learning models for recognizing ASL gestures from video clips using the WLASL dataset. Our approach combined spatial feature extraction using a convolutional neural network with temporal modeling using an LSTM network. The goal of this work was to evaluate whether incorporating temporal information improves generalization performance compared to purely spatial models.

Across our experiments, the CNN-LSTM architecture produced the strongest overall performance, achieving the best validation accuracy when evaluated using signer-based dataset splits. While CNN-only models achieved very high training accuracy, they exhibited clear signs of overfitting and large generalization gaps between training and validation results. By incorporating temporal modeling, the CNN-LSTM model was better able to capture motion patterns across frames, which improved performance on previously unseen signers. These results highlight the importance of temporal representation learning for video-based sign language recognition.

In terms of our original project goals, we aimed to design and evaluate a deep learning pipeline capable of recognizing ASL glosses from video sequences. We successfully implemented multiple architectures, explored preprocessing strategies such as region of interest cropping and temporal frame sampling, and evaluated the

effect of signer-based dataset splits. However, several planned improvements were not fully explored due to time and computational constraints. In particular, we did not extensively address class imbalance across gloss categories or experiment with more advanced video architectures such as 3D convolutional networks or transformer-based models. Future work could build upon our results by incorporating these techniques, expanding the vocabulary size, and further improving robustness across different signers and environments.

VIII. CONTRIBUTIONS

Matthew contributed ideas and feedback during the early ideation phase of the project. In the following project stages, he added graphical and tabular explorations of the dataset, cleaned up the project notebook, and assisted with the implementation of CV2 and bounding boxes. During the model development and testing phases, Matthew created the machine learning pipeline and provided us with the framework that our model was built upon. He also developed the backbone of the ASL project report.

Jaron helped refine the scope and strategy of the project during its early stages and edited the capstone abstract. He also implemented a baseline system to process ASL videos using bounding boxes (BBox), oversaw the integration pipeline, and began the development process of dividing videos into frames. During the model development phase, Jaron also refined and fine tuned the model to achieve a high validation score. This included fine tuning parameters, restructuring the pipeline, and introducing new variables such as learning rate.

Franky contributed to drafting the abstract and provided insights on different modeling approaches during team meetings. He also imported and cleaned the dataset, refined graphical and tabular data exploration, and supported the implementation of bounding boxes (BBox). In addition, he created presentation slides and filled it with relevant information and visualizations. Lastly, he improved the model's parameters and optimized run efficiency, as well as contributed to expanding and refining the ASL project report.

IX. REFLECTION

In hindsight, there are several aspects of this project that we would approach differently if given the opportunity to repeat it. First, selecting the dataset was more challenging than expected as the initial version we chose had almost half of the videos missing, which caused delays and required us to switch to a more complete version. If we were to redo the project, we will start by thoroughly vetting the dataset on its completeness and quality to avoid setbacks.

One of the main challenges we encountered was the complexity and scale of the WLASL dataset. Handling video data required significant preprocessing and computational resources, which limited the number of experiments we were able to perform. If we were starting the project again, we would spend more time early on building a more efficient data pipeline and verifying dataset integrity before beginning model training.

Another area we would improve is the exploration of alternative modeling approaches. While our project focused primarily on CNN and CNN–LSTM architectures, modern video recognition research increasingly uses architectures such as 3D convolutional networks or transformer-based models. Exploring these models may have led to improved performance and deeper insights into how temporal motion patterns contribute to sign recognition.

Finally, we would devote more effort to addressing class imbalance within the dataset. Many ASL glosses appear far less frequently than others, which can bias model training and reduce accuracy for rare signs. Techniques such as weighted loss functions or data resampling could potentially improve performance in future work. Despite these challenges, the project provided valuable experience working with large-scale video datasets and developing deep learning pipelines for complex visual recognition tasks.

X. REFERENCES

[1] Li, D., et al., Word-Level Deep Sign Language Recognition.

[2] WLASL Dataset.

[3] He, K., et al., Deep Residual Learning for Image Recognition.

[4] Hochreiter, S., Schmidhuber, J., Long Short-Term Memory.

[4] Code libraries used: torch, torch vision, numpy, pandas, matplotlib, cv2

XI. RESPONSE TO PEER REVIEWS

We appreciate the thoughtful feedback provided by Groups 11 and 15 and have incorporated several of their suggestions into the final version of our report.

First, both review groups noted the clarity of our experimental walkthrough and the usefulness of visual examples in demonstrating the preprocessing pipeline. We are glad that these aspects were helpful and have retained the dataset visualizations and preprocessing illustrations to maintain clarity in the explanation of our approach.

Several reviewers suggested that additional evaluation visualizations would strengthen the analysis. In particular, both groups recommended including a confusion matrix to better understand which ASL glosses were most frequently misclassified. In response, we expanded the discussion of model evaluation and included additional analysis of model performance to better explain the behavior of the classifier and the types of errors that may occur when signs share similar visual features.

Another comment raised by Group 11 was the absence of traditional baseline comparisons for the deep learning models. While the primary goal of our work was to compare CNN and CNN–LSTM architectures for temporal modeling, we clarified this experimental focus in the report and expanded the discussion of why these architectures were selected for evaluation.

Group 15 also highlighted the potential overfitting observed in the models due to the large gap between training and validation accuracy. We acknowledge this limitation and have expanded the discussion of overfitting in the experiments section. In particular, we now discuss how the CNN–LSTM architecture reduces the training–validation gap and how future work could incorporate additional regularization methods such as dropout, stronger data augmentation, or alternative architectures.

Finally, Group 15 suggested that landmark-based representations might offer a more computationally efficient alternative to processing raw video frames. While our current work focuses on appearance-based video models, we agree that pose or landmark-based approaches are promising directions for future research and acknowledge this as a potential extension in the discussion of future work.

Overall, we appreciate the constructive feedback from both groups. Their suggestions helped us improve the clarity of our evaluation and strengthen the discussion of model behavior and limitations in our final report.